

PROJECT PROGRESS LOG

MOBILE CHARGER

Team Members

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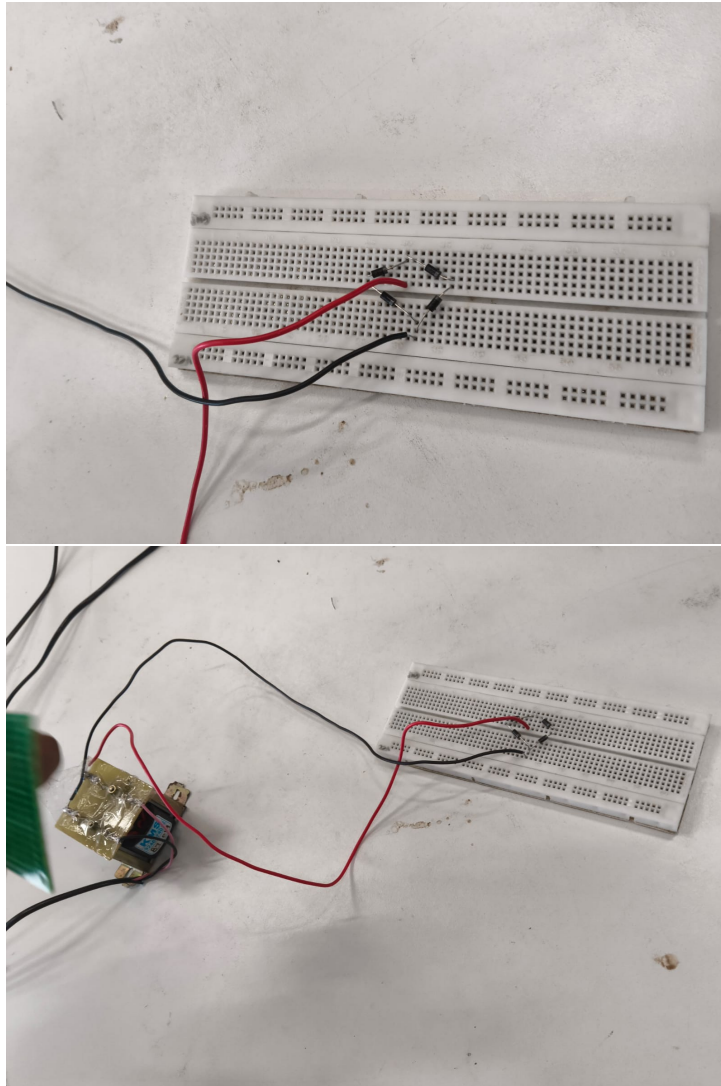


Figure 1: Initial arrangement/Verification

0.1 Work Done(Figure: 1)

We first arranged the diodes on the breadboard (as referred in main.pdf).The connections were made to the breadboard and the transformer.We checked the components before working on the soldering part of the components to the circuit.

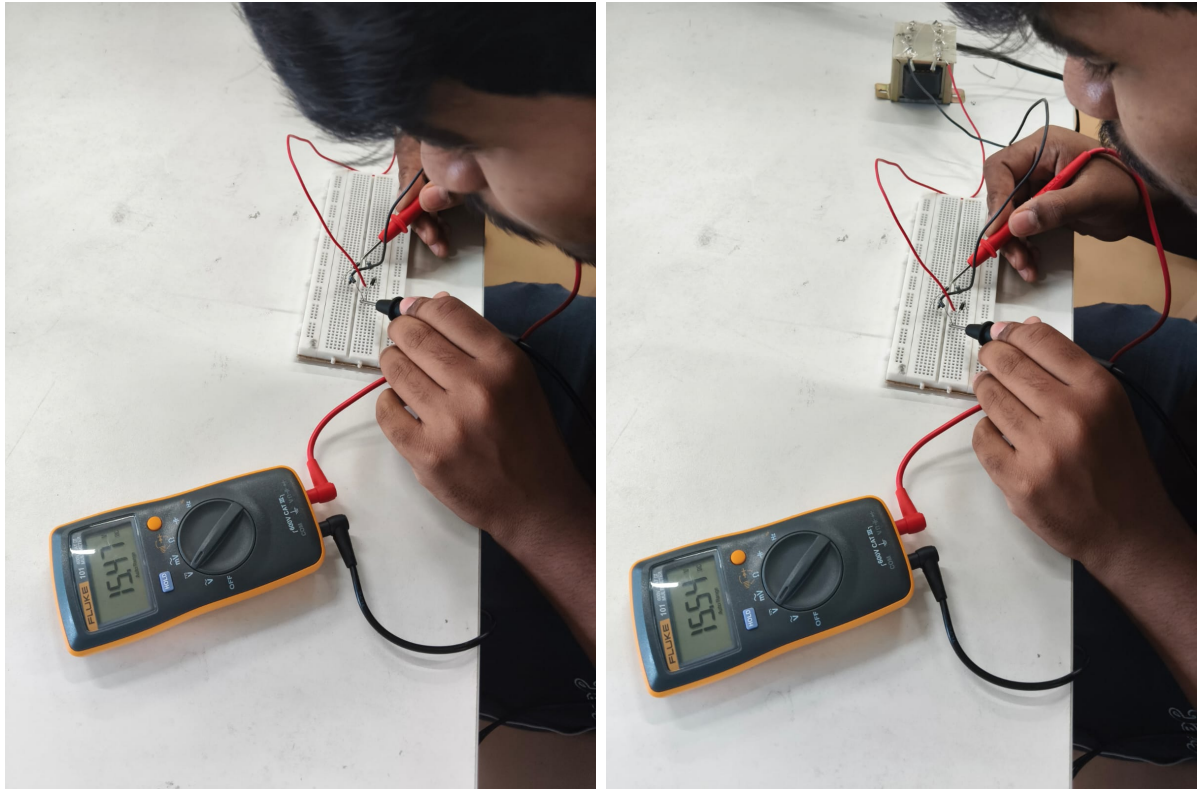


Figure 2: Enter Caption

0.2 Work Done(Figure: 2)

After giving power to the transformer, we checked the output with the help of a multimeter.

0.3 Observation: (Figure: 2)

The readings were not accurate. So we took a few readings of the output voltage. By these readings of the multimeter, we observed that the DC voltage showing in the multimeter is between 15.4V and 15.6V

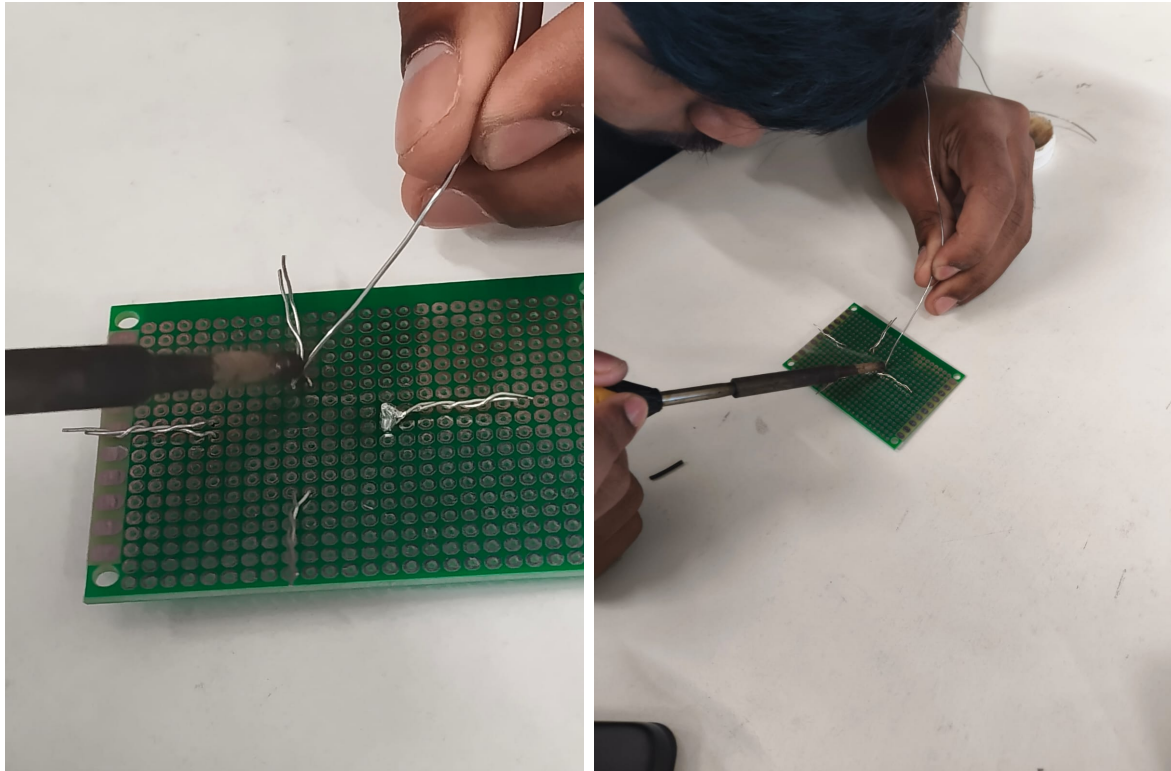


Figure 3: Arrangement of Diodes on the PCB

0.4 Work Done (Figure: 3)

We then arranged the diodes on the PCB and twisted the ends of the diode as shown in Figure: 3. We soldered each of the two diodes as shown in Figure: 3.

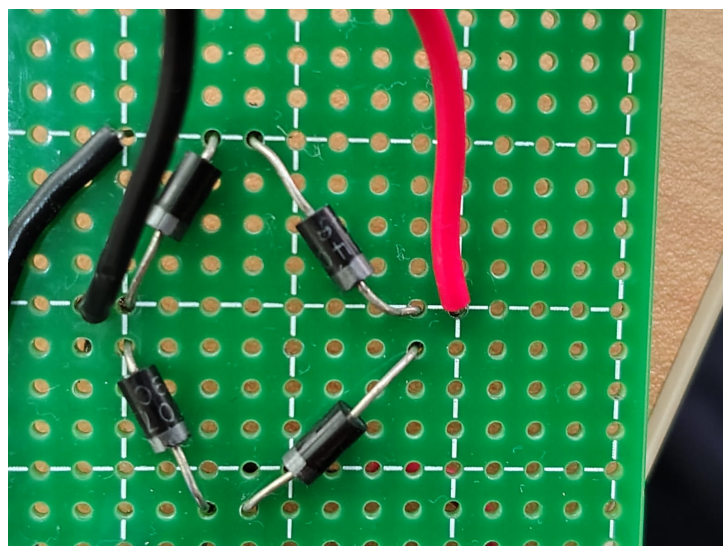


Figure 4: The view of the circuit from above

0.5 Work Done (Figure 4)

Then we made connections to the diodes and the transformer on the PCB.

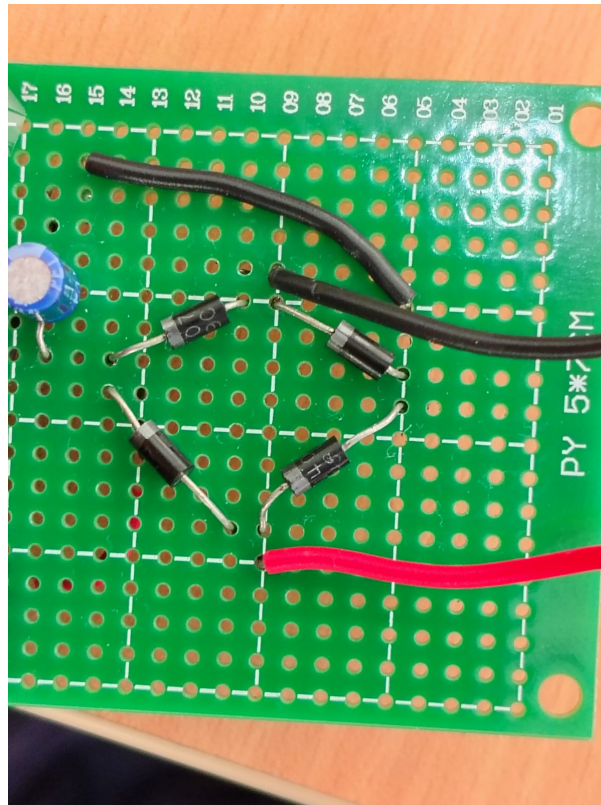


Figure 5: The circuit after connecting capacitor

0.6 Work Done (Figure: 5)

Connection was made to a capacitor of $100\ \mu\text{F}$ to the soldered diodes on the PCB and connected the negative terminal of the capacitor to the ground of the transformer and soldered all the connections made to the capacitor.

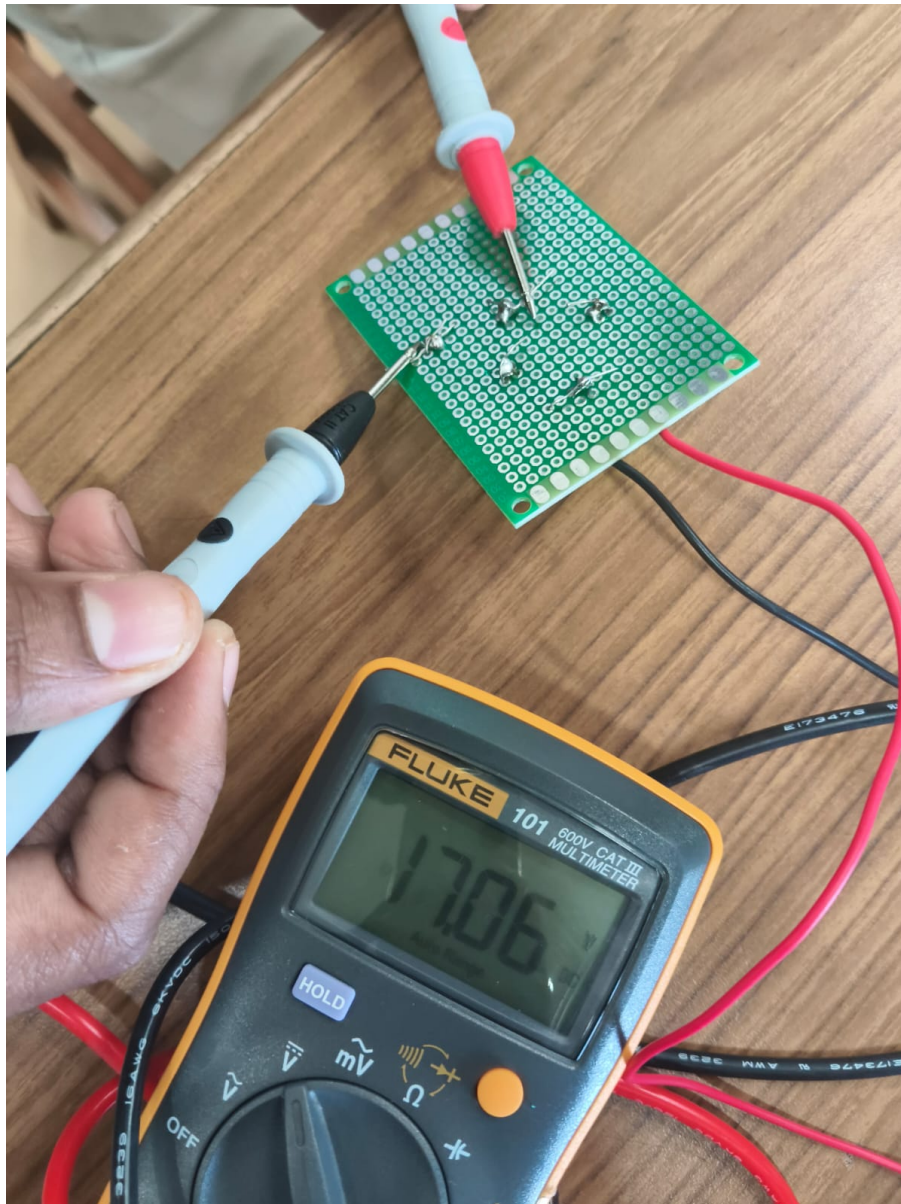


Figure 6: Checking the output voltage of the capacitor

0.7 Observations (Figure: 6)

After soldering, we checked the output voltage of the 100 μ F capacitor with the help of a multimeter. The readings were not accurate again. It has given a output voltage of approximately 17V to 18V.

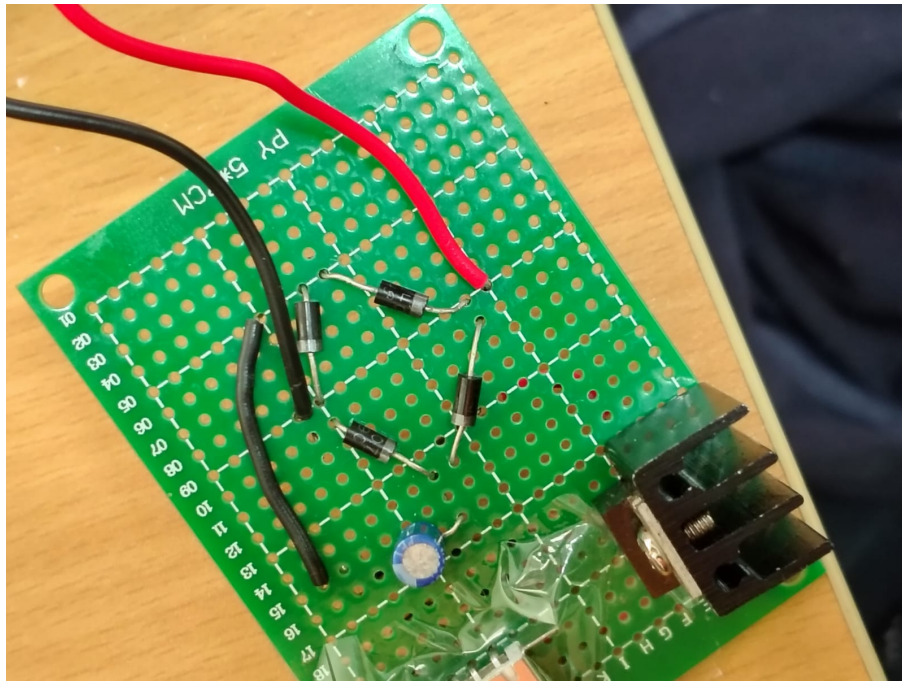


Figure 7: Circuit after connecting voltage regulator

0.8 Work Done (Figure: 7)

we connected the voltage regulator to the PCB as shown in Figure 7, Then firmly soldered the connections on the underside of the PCB.

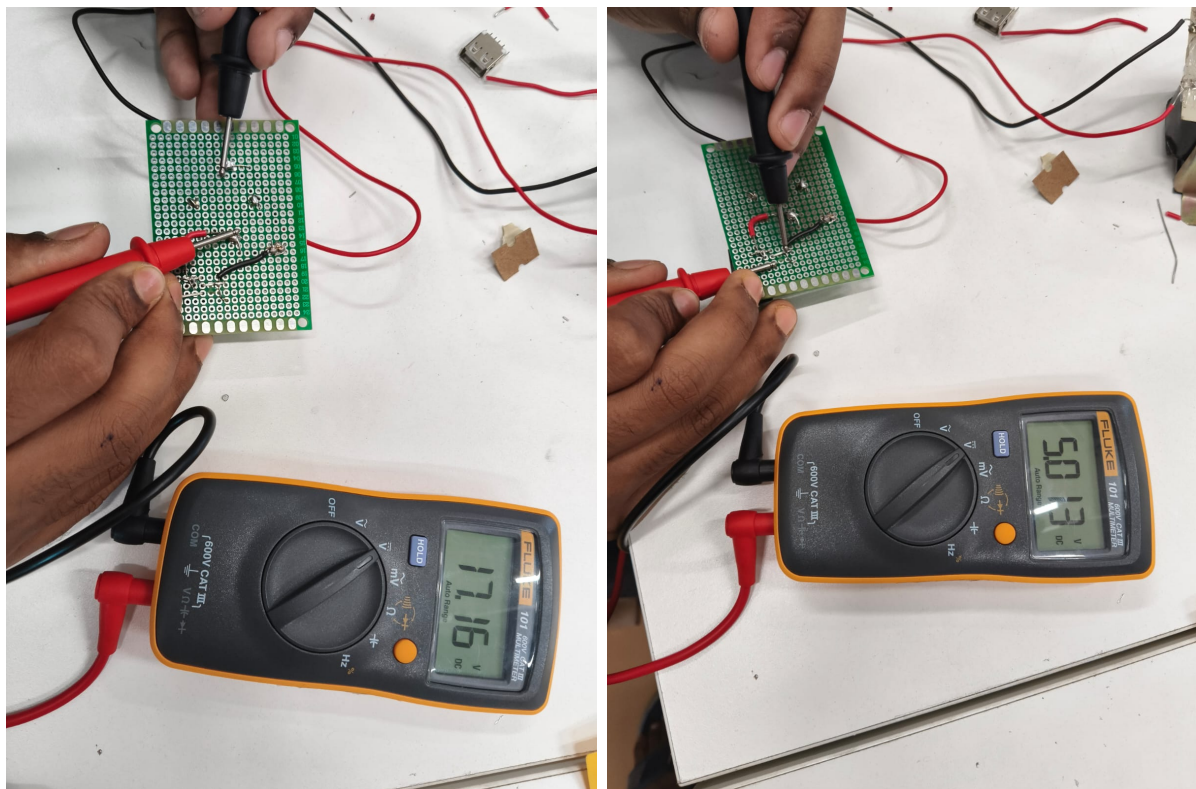


Figure 8

0.9 Observation (Figure 8)

We verified both the output voltages of diodes and the voltage regulator after firm soldering with the help of a multimeter. We observed that the input voltage to the regulator is around 5V. The output voltage of the diodes remained same.

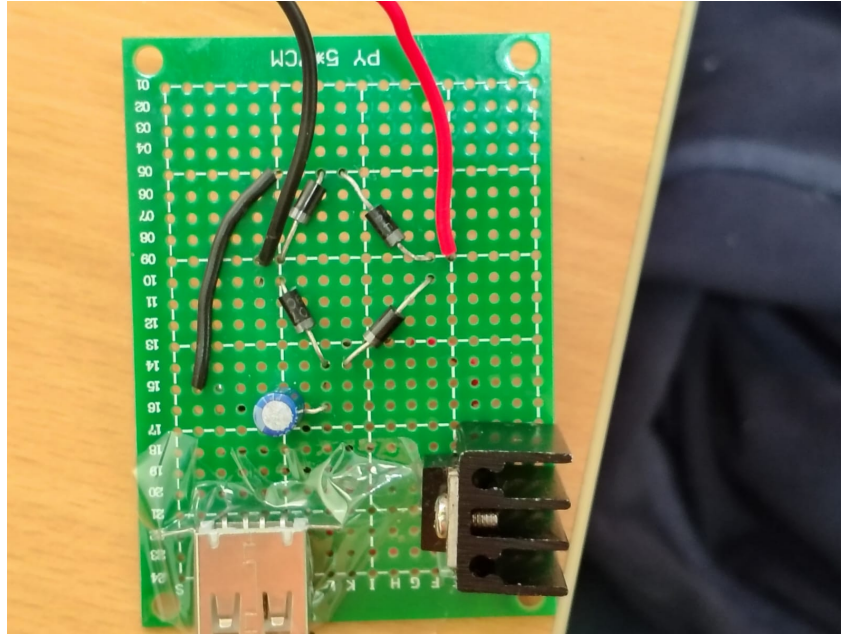


Figure 9: PCB after connecting USB port

0.10 Work Done (Figure 9)

We bent the pins of the 4 pins of the USB to place it on the PCB horizontally. Then we made connections to the ground and the output pin of voltage regulator with wired and them we soldered the connections made to the USB

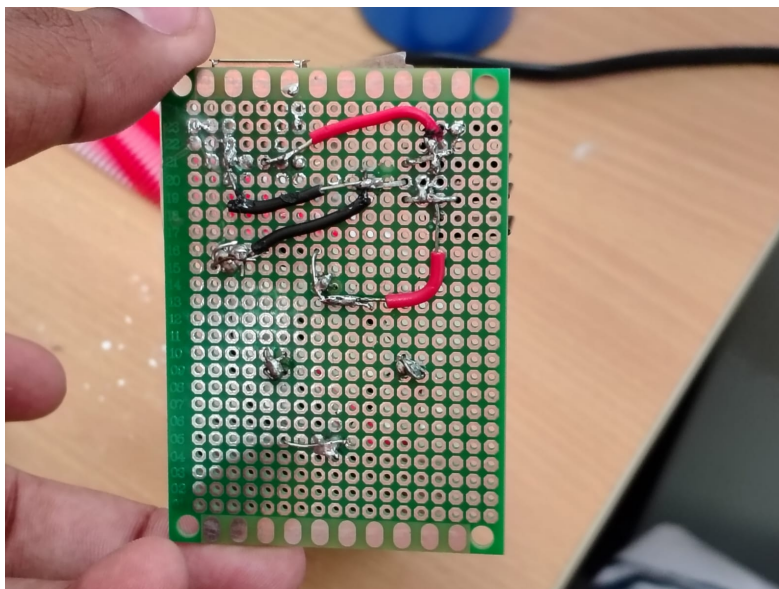


Figure 10: The soldered side of the PCB board

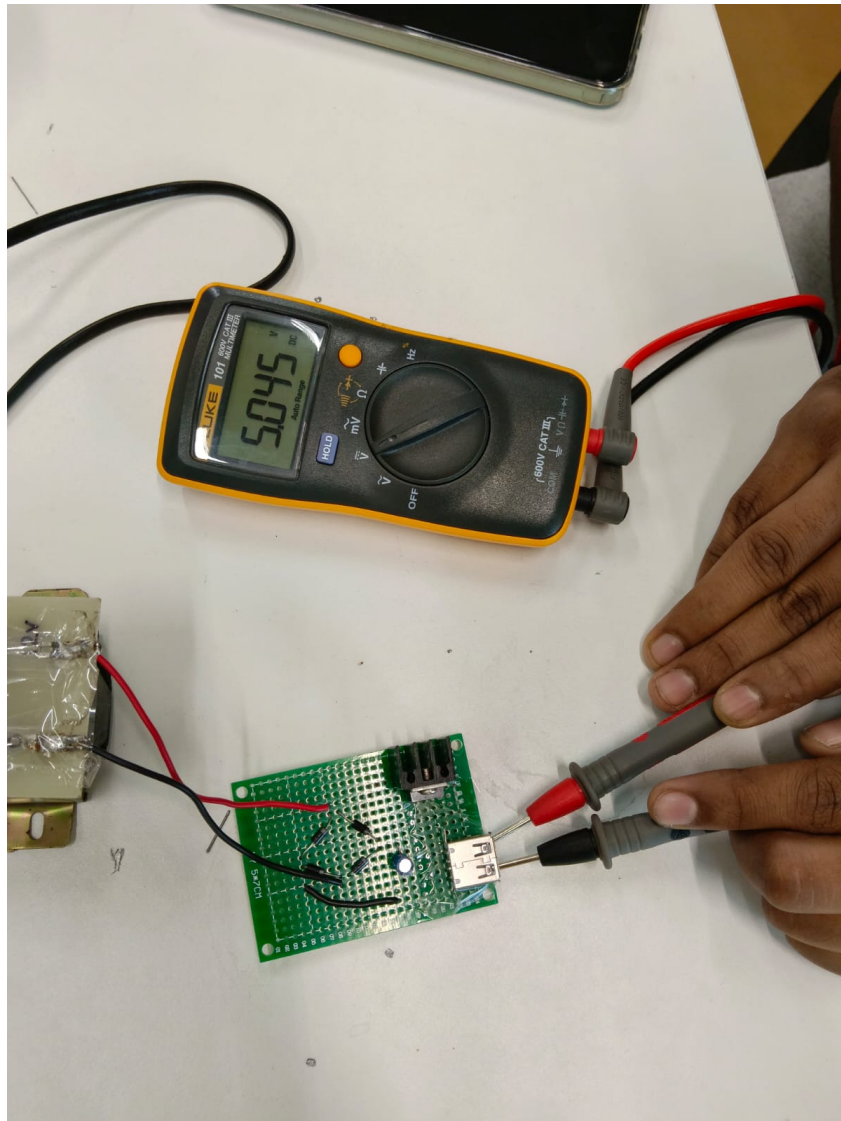


Figure 11: Verifying the output voltage of the USB

0.11 Observation (Figure 11)

We observed the output DC voltage of the USB with the help of a multimeter. The voltage we observed with the help of a multimeter is approx 5V. It was not exactly 5V. It gave more than 5V, as shown in Figure 11

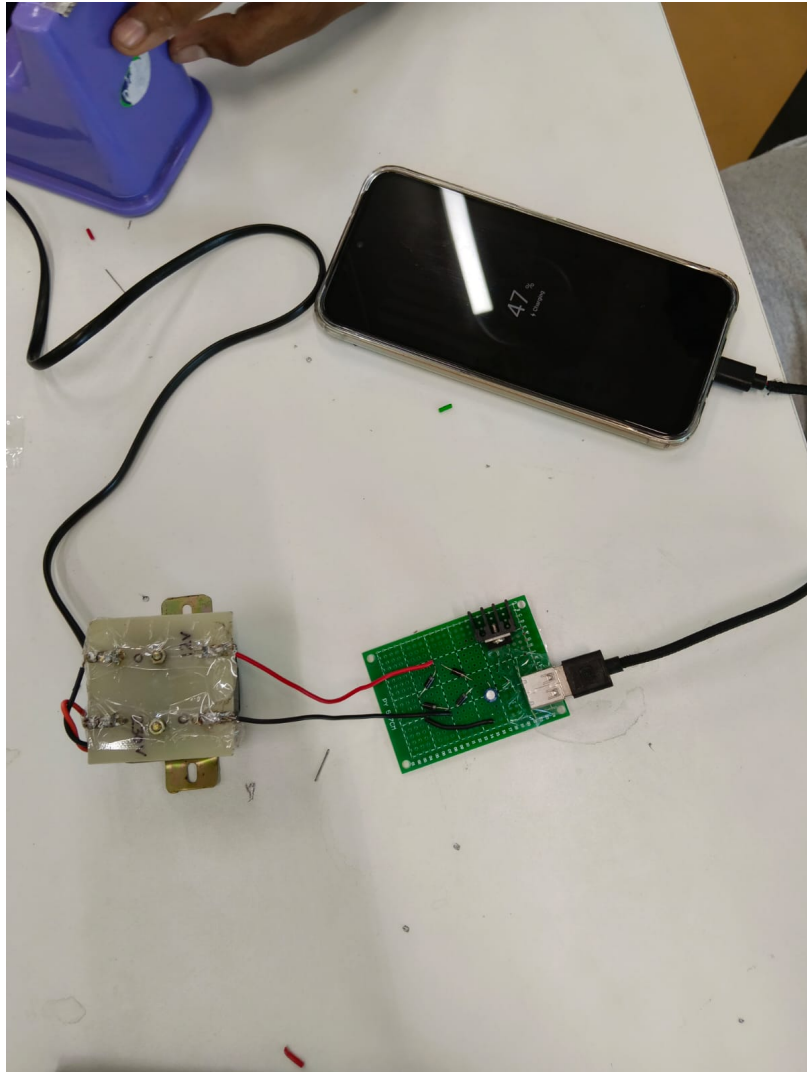


Figure 12: Verification the product

0.12 Verification (Figure 12)

We verified the product by taking cable consisting of Male USB pin and Male C pin. The Male USB pin is plugged in the Female port on the PCB. The C pin is plugged into our mobile. As shown in the Figure 12 the mobile was being charged.

0.13 3D Model

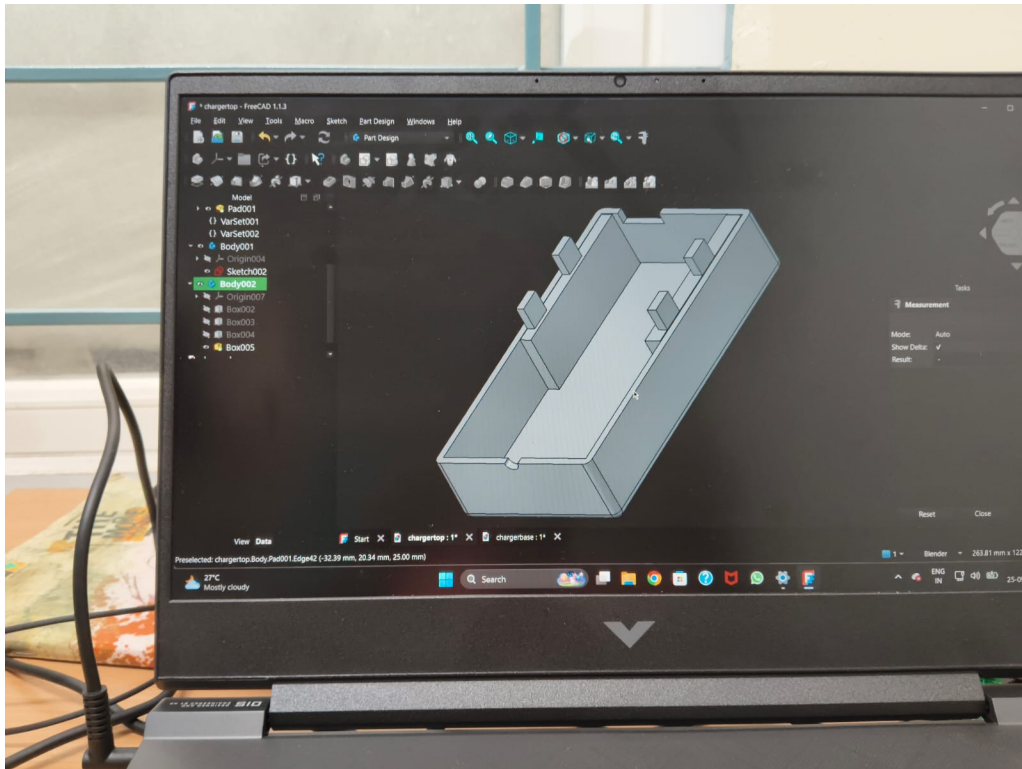


Figure 13: 3D Model of the Top part of the case

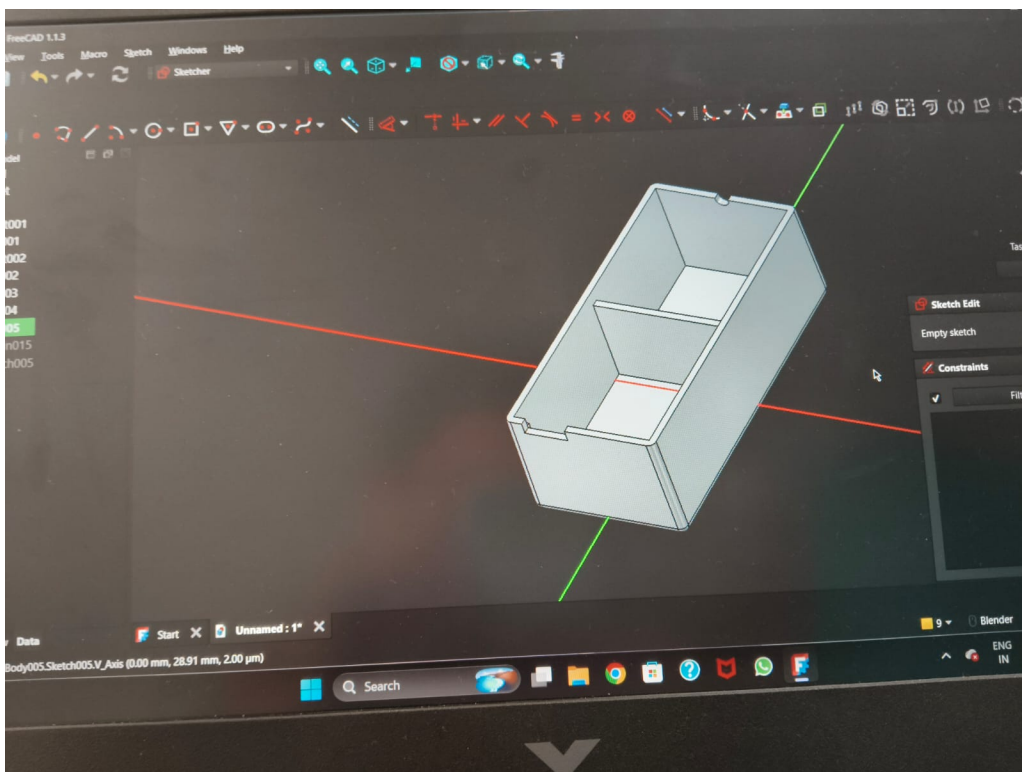


Figure 14: 3D Model of the Bottom part of the case

Figure 13 and Figure 14 are the 3D Models we prepared as a case for our charger. One is the top part and other is the bottom part. We thought of encasing the PCB and the transformer in the case we created.